THE 40 HOUR CABIN · MODULAR HOUSING · BY A VETERAN, FOR VETERANS

Efficiency, by the numbers

One standard body against the houses it would replace.

this building · the comparison · variant or upkeep

PRICE A SQUARE FOOT

New site-built house, without land **$217**

New manufactured home **$87**

This standard building **$43.45**

The standard building is 50% of a new manufactured home's price a square foot, margin included and the pad extra. Not like for like: one room with a service core, not a finished house.

MODELLED · PUBLISHED (NAHB, CENSUS)

WHERE THE $12,036 GOES



- Frame **$692** 6%
- Weather shell **$3,265** 27%
- Utility column **$1,405** 12%
- Services **$485** 4%
- Utility panels **$491** 4%
- Build labour **$1,884** 16%
- Overhead and warranty **$1,809** 15%
- Maker's margin **$2,006** 17%

The core is $1,890, 30% of the build. It lasts 25 years and moves to the next dome for $260.

MODELLED

OUTSIDE SKIN, SAME FLOOR

Box house **997 sq ft**

Dome **583 sq ft**

42% less to build, to weatherproof, and to lose heat through.

MODELLED

HEATING AND COOLING, A YEAR

Box house **$220**

Dome **$129**

$91 a year never spent, from the shape alone.

MODELLED

WIND LOAD

Box house, drag **1.05**

Dome, drag **0.42**

60% less push from the same wind: the shape sheds it.

PUBLISHED COEFFICIENTS

RAIN THE SEAMS CATCH, A YEAR

20 in of rain a year **4,935 gal**

40 in of rain a year **9,870 gal**

60 in of rain a year **14,805 gal**

550 sq ft of shell, 72% caught: 247 gallons for every inch that falls, run by the seams into the pad's tank. No gutters.

MODELLED · RAINFALL DECLARED

ONE PARTS LIST, THREE SIZES

10 ft across **79 sq ft**

19.4 ft across **296 sq ft**

30 ft across **707 sq ft**

Every size is the same 120 members and 40 panels on the same two lengths: one jig, one hardware set, one way of working.

MODELLED

THE SKIN, AND KEEPING IT

Woven sailcloth **$2,809**

Cotton duck + silicone **$1,486**

Beeswax recoat **$34**

2 layers, 112 m² of 100% cotton soaked in silicone. A beeswax coat every 1–2 years: 2.2 kg, about 6 hours.

ESTIMATED · SEE THE TABLE

WOOD KEPT FROM THE LOG

Sawn into two-by-fours **45%**

A good mill, mixed sizes **60%**

Split into wedges **88%**

The frame comes from the tree where it stands; the corners of a log are not waste.

SIMULATED · MILL ALLOWANCE THE AUTHOR'S

SNAP-IN, NOT BUILT-IN

Flue panel **$180**

Mirror panel **$275**

Skylight panel **$395**

Open-bay door **$1,480**

12 panel types and 25 snap-in modules, from $180 to $1,480. A bay changes with a driver, not a builder.

MODELLED

THE CORE, OVER ITS LIFE

Build a core **$1,890**

Move it to the next dome **$260**

Each year of its 25 **$76**

Buy the plumbing once. The core carries 5 services (air, drain, power, structure, water) and outlives the shell around it.

MODELLED

HOURS OF WORK

Splitting, measured **16 h**

Frame, mistakes allowed for **37.5 h**

Whole building **67 h**

The frame clock keeps 7.5 hours for getting it wrong. The whole building, counted the same way, is about 67 hours.

MEASURED · ESTIMATED

DECLARED, NOT COMPUTED: THE BEESWAX UPKEEP

wax_g_m2	40 g/m2	estimate	a maintenance coat rubbed into cloth already proofed with silicone: far less than a first waxing of bare canvas
wax_usd_kg	15 $/kg	estimate	beeswax pellets bought by the kilogram
wax_m2_hour	10 m2/hour	estimate	rubbing in and melting off with a heat gun
wax_every_lo	1 years	owner's practice	recoat yearly where it is wet and sunny
wax_every_hi	2 years	owner's practice	every second year where it is not

Every bar drawn to scale from the project's own models on 13 October 2026. Sources and code: **github.com/dkzeanah/DomeSimulator**. Modelled figures are what the software computes; published ones are cited in its source; estimates are declared above or in the module that uses them.